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Introduction

Over the last few years, digital payments have become a regular part of our daily lives. In India especially, UPI has completely changed the way people send and receive money. From paying for groceries to splitting bills with friends, everything can now be done in seconds. But along with this convenience, there has also been a noticeable rise in fraud cases.

Many people fall into traps like fake payment requests, phishing messages, or calls from scammers pretending to be bank officials. What makes this more dangerous is that these frauds happen instantly, leaving very little time to react.

In this project, I have tried to explore how Machine Learning can help in identifying such fraudulent transactions. Instead of manually checking each activity, the system learns patterns from past data and tries to predict whether a transaction looks suspicious or not. The goal is not to build a perfect system, but to understand how AI can be applied to solve real-world problems in a practical way.

Objectives

- To study different types of UPI-related frauds happening in real life
- To understand how machine learning models detect patterns in data
- To build a simple but meaningful fraud detection system
- To evaluate how accurate the model is
- To explore improvements that can be made in the future

Problem Statement

With the increasing dependency on UPI for digital transactions, fraud cases are also rising rapidly. Most current systems depend heavily on user awareness, which is not always reliable. There is a need for an intelligent system that can automatically analyse transaction behaviour and detect suspicious activities before damage is done. This project attempts to address this problem using machine learning techniques.

Types of UPI Fraud

Some common types of fraud include:

- **Phishing Attacks:** Users receive fake links that look like official payment pages
- **Collect Request Scams:** Fraudsters send a “request money” notification instead of payment
- **QR Code Fraud:** Scammers trick users into scanning fake QR codes
- **Impersonation Calls:** Pretending to be bank officials or customer support
- **Screen Sharing Scams:** Asking users to install apps that give remote access

What’s common in all these is **manipulation + urgency**, which leads users to act without thinking.

Machine Learning is a branch of Artificial Intelligence where systems learn patterns from data instead of being explicitly programmed.

In this project, we use **Supervised Learning**, where:

- The model is trained on labelled data
- Each transaction is marked as **fraud (1)** or **not fraud (0)**

Why Classification?

Because we are predicting categories:

- Fraudulent
- Genuine

Why Decision Tree?

- Easy to understand and explain
- Works well for small datasets
- Visual decision-making process

Decision trees split data based on conditions like:

- “Is amount > 5000?”
- “Is transaction at odd hours?”

System Architecture

1. Data Collection
2. Data Preprocessing
3. Feature Selection
4. Model Training
5. Prediction
6. Result Display

This flow shows how raw transaction data becomes a meaningful prediction.

Working of the System

When a user makes a transaction, certain details like amount, time, and receiver information are considered. These details are passed into the trained model. The model compares this data with patterns it has already learned.

For example, if a transaction is unusually large and happens late at night to an unknown account, the model may flag it as suspicious. Based on this, the system outputs whether the transaction is safe or potentially fraudulent.

Implementation

Code:

```
# UPI Fraud Detection System

import pandas as pd
import numpy as np
from sklearn.model_selection import train_test_split
from sklearn.tree import DecisionTreeClassifier
from sklearn.metrics import accuracy_score, classification_report, confusion_matrix


# Step 1: Load Dataset

try:
    data = pd.read_csv("upi_fraud_dataset.csv")
    print("Dataset loaded successfully.\n")
except:
    print("Dataset not found. Creating sample dataset...\n")

    # Creating a sample dataset
    data = pd.DataFrame({
        'amount': [100, 5000, 200, 7000, 150, 9000, 300, 12000, 250, 8000],
        'time': [10, 23, 14, 1, 9, 2, 16, 0, 11, 3],
        'is_new_receiver': [0, 1, 0, 1, 0, 1, 0, 1, 0, 1],
        'multiple_txn': [0, 1, 0, 1, 0, 1, 0, 1, 0, 1],
        'fraud': [0, 1, 0, 1, 0, 1, 0, 1, 0, 1]
    })


# Step 2: Data Preprocessing

print("First 5 rows of dataset:\n", data.head(), "\n")

X = data[['amount', 'time', 'is_new_receiver', 'multiple_txn']]
y = data['fraud']

# Step 3: Train-Test Split

X_train, X_test, y_train, y_test = train_test_split(
    X, y, test_size=0.2, random_state=42
)
```

```python
# Step 4: Model Training

model = DecisionTreeClassifier(max_depth=4, random_state=42)
model.fit(X_train, y_train)

print("Model training completed.\n")

# Step 5: Prediction
y_pred = model.predict(X_test)

# Step 6: Evaluation
accuracy = accuracy_score(y_test, y_pred)

print("Model Accuracy:", accuracy)
print("\nClassification Report:\n", classification_report(y_test, y_pred))
print("\nConfusion Matrix:\n", confusion_matrix(y_test, y_pred))

# Step 7: Manual Testing (User Input)
print("\n--- Test a New Transaction ---")

try:
    amount = float(input("Enter transaction amount: "))
    time = int(input("Enter transaction hour (0-23): "))
    new_receiver = int(input("Is new receiver? (0=No, 1=Yes): "))
    multiple_txn = int(input("Multiple transactions recently? (0=No, 1=Yes): "))

    user_data = np.array([[amount, time, new_receiver, multiple_txn]])
    result = model.predict(user_data)

    if result[0] == 1:
        print("\n⚠ Warning: This transaction may be FRAUDULENT!")
    else:
        print("\n✅ This transaction appears SAFE.")

except:
    print("Invalid input. Please try again.")
```

Result:

```
IDLE Shell 3.14.0
File  Edit  Shell  Debug  Options  Window  Help
========================== RESTART: C:/Users/rutuj/AppData/Local/Programs/Python/Python314/UPI Fraud Detection (Code).py ==========================
Dataset not found. Creating sample dataset...

First 5 rows of dataset:
    amount  time  is_new_receiver  multiple_txn  fraud
0      100    10                0             0      0
1     5000    23                1             1      1
2      200    14                0             0      0
3     7000     1                1             1      1
4      150     9                0             0      0

Model training completed.

Model Accuracy: 1.0

Classification Report:
               precision    recall  f1-score   support

           0       1.00      1.00      1.00         1
           1       1.00      1.00      1.00         1

    accuracy                           1.00         2
   macro avg       1.00      1.00      1.00         2
weighted avg       1.00      1.00      1.00         2


Confusion Matrix:
 [[1 0]
 [0 1]]

--- Test a New Transaction ---
Enter transaction amount: 3000000
Enter transaction hour (0-23): 10
Is new receiver? (0=No, 1=Yes): 1
Multiple transactions recently? (0=No, 1=Yes): 0

Warning (from warnings module):
  File "C:\Users\rutuj\AppData\Local\Programs\Python\Python314\Lib\site-packages\sklearn\utils\validation.py", line 2691
    warnings.warn(
UserWarning: X does not have valid feature names, but DecisionTreeClassifier was fitted with feature names
```

Results & Analysis

After training the model, we test it using unseen data. The model gives predictions based on patterns it has learned. In most cases, it correctly identifies suspicious transactions, especially when the features clearly indicate unusual behaviour.

The accuracy may vary depending on the dataset, but even with a simple dataset, the model demonstrates how machine learning can be applied in fraud detection.

Advantages

- Helps in early detection of fraud
- Reduces dependency on manual monitoring
- Can be scaled with larger datasets
- Provides quick decision-making

Limitations

- Small dataset limits performance
- Real-world fraud patterns are more complex
- Requires continuous updates
- Cannot replace human judgment completely

Future Scope

- Use real transaction datasets (if available securely)
- Apply advanced models like Random Forest or Neural Networks
- Integrate with mobile payment apps
- Add real-time alerts and notifications
- Combine with user behaviour analytics

Conclusion

While working on this project, I realized how common digital fraud has become and how important it is to have smart systems to deal with it. UPI has made life easier, but it also comes with certain risks that cannot be ignored.

Through this project, I understood the basic idea of how machine learning models can identify patterns and make decisions based on data. Even though the model used here is simple, it still gives a good idea of how fraud detection systems work in real life.

This project is a small step towards understanding how AI can be used to make digital systems safer. With more data and better models, such systems can become even more accurate and reliable in the future.

References

- Scikit-learn Documentation
- Python Official Documentation
- Articles on UPI fraud awareness (RBI guidelines, blogs)
- Research papers on fraud detection using machine learning